AI Chatbot & AI Agent

Intelligent AI Customer Support & Product Discovery Platform

Project Type	AI-Powered Customer Support & Sales Assistant
Industry	Industrial Machinery / Product Sales
Role	Full Stack MERN Developer & AI Integration Developer

AI & LLMs	OpenAI, Anthropic Claude, Google Gemini
AI Architecture	AI Agents, RAG, Semantic Search, Vector Embeddings, Knowledge Base
Backend	Node.js, Express.js, REST APIs
Database	MongoDB, Mongoose
Frontend	React.js
Integrations	Product APIs, Knowledge Base, Email, Lead Management
Infrastructure	DigitalOcean, Nginx, PM2, GitHub Actions

Project Overview

The AI Chatbot & AI Agent was developed as an intelligent customer-facing assistant for an industrial machinery business.

Instead of relying on a traditional FAQ chatbot with predefined responses, the system uses modern Large Language Models, structured product information, and a knowledge base to understand customer questions and generate contextual responses.

The assistant was designed to work as a combination of:

AI Customer Support + Product Discovery Assistant + Recommendation Engine + Lead Qualification Agent

Customers can ask questions naturally, explore machinery, request product information, receive recommendations, and move through the purchasing journey without requiring immediate human assistance.

The Problem

Industrial machinery customers often have highly specific questions before making an enquiry. For example:

- Which machine is suitable for my requirement?

- What is the capacity of this forklift?
- Which attachment works with this machine?
- What is the difference between two models?
- Is this machine suitable for a particular application?
- What attachments are available?
- Can I get more information about this product?
- How can I contact the company?

A traditional website search or static FAQ system cannot handle these questions effectively. The goal was therefore to build an AI assistant capable of understanding natural-language questions and connecting them with the company's actual product and business knowledge.

Solution

The resulting platform provides an AI-powered conversational interface where customers can interact naturally with the business. Instead of forcing customers to navigate multiple pages, the assistant can guide them through product discovery.

```
Customer Question
       |
AI Understanding
       |
Intent Detection
       |
Knowledge / Product Retrieval
       |
LLM Reasoning
       |
Contextual Response
       |
Recommendation / Next Action
```

This creates a more natural customer experience than traditional keyword-based search.

1. Intelligent Customer Support

The chatbot acts as a first-line customer support assistant. Customers can ask questions in natural language instead of searching through documentation manually. For example:

> *“I need a forklift for heavy warehouse operations. What would you recommend?”*

The AI analyzes the request and uses available product information to provide a relevant response. It can also answer common questions about:

- Products
- Specifications
- Features
- Attachments
- Applications
- General FAQs

- Business information
- Purchasing process
- Customer enquiries

2. Product Discovery

One of the main purposes of the assistant is helping customers discover the right machinery. Instead of simply searching for a product name, customers can describe what they need.

```
Customer
   |
"I need a forklift for a warehouse
 where I regularly move heavy pallets."
   |
   v
   AI
   |
Understands requirements
   |
Searches product knowledge
   |
Identifies relevant machines
   |
Provides recommendations
```

This turns the chatbot into a conversational product discovery system.

3. AI Product Recommendations

The assistant can use customer requirements to recommend relevant products. The recommendation process considers available product information and the context of the conversation.

```
Customer Requirements
        |
Application / Capacity /
Environment / Attachment Reqs
        |
Product Knowledge
        |
Candidate Products
        |
AI Reasoning
        |
Recommended Machine
```

This allows customers to start with a business requirement rather than already knowing which product they want.

4. Knowledge Base Integration

A major component of the system is its knowledge base. Instead of expecting the LLM to know every business-specific detail, relevant information is stored in a structured knowledge system, including:

- Product information
- Machine specifications
- Features

- FAQs
- Attachments
- Business information
- Product documentation
- Support information
- Other company-specific content

This gives the AI access to information specific to the business.

5. RAG Architecture

The chatbot uses a Retrieval-Augmented Generation (RAG) approach. Instead of sending every question directly to an LLM, the system first retrieves relevant information from the available knowledge. The basic architecture is:

```
User Question
      |
Text Processing
      |
Embedding Generation
      |
Semantic Search
      |
Relevant Knowledge
      |
LLM
      |
Context-Aware Answer
```

This allows the model to generate responses based on business-specific information.

6. Semantic Search

Traditional keyword search may fail when customers use different terminology. For example:

```
Customer:
"machine for moving heavy warehouse loads"

Knowledge:
"high-capacity warehouse forklift"
```

The wording is different, but the meaning can be related. Semantic search helps the system identify relevant information based on meaning rather than exact keyword matching. This makes the assistant much more useful for natural-language customer conversations.

7. Vector Embeddings

Knowledge-base content can be transformed into vector embeddings. Conceptually:

```
Product Information
       |
Text Chunking
       |
Embedding Model
       |
Vector Representation
       |
```

```
Vector Search
```

When a customer asks a question, the query is also converted into an embedding. The system then searches for semantically similar knowledge before passing the relevant context to the LLM.

8. Multi-LLM Architecture

The platform was designed to work with multiple major LLM providers.

Supported Providers

OpenAI — used for powerful general-purpose conversational AI and product assistance.

Anthropic Claude — used for advanced reasoning and business-oriented AI workflows.

Google Gemini — used as another LLM provider for conversational and AI capabilities.

3 Major LLM Providers

This makes the architecture more flexible than building the entire system around a single AI provider.

9. AI Agent Architecture

Beyond simple question-and-answer functionality, the system also includes AI Agent workflows. The difference is:

```
Traditional Chatbot
Question -> Answer

AI Agent
Question
   |
Understand Intent
   |
Determine Required Action
   |
Retrieve Information
   |
Process Context
   |
Generate Response
   |
Perform / Trigger Workflow
```

This allows the assistant to behave more like an intelligent business representative.

10. Conversational Context

The chatbot maintains conversation context so customers do not need to repeat the same information in every message. For example:

```
Customer:
"I need a forklift for warehouse use."

AI:
"What lifting capacity are you looking for?"

Customer:
```

```
"Around 3.5 tonnes."

AI:
"Based on your requirement, I can help
 you explore suitable 3.5-tonne machines."
```

The second response can use information provided earlier in the conversation. This creates a more natural conversation flow.

11. Intent Understanding

The assistant can identify what the customer is trying to accomplish. Potential intents include:

- Product Search
- Product Recommendation
- Product Question
- Specification Request
- Attachment Question
- FAQ
- Sales Enquiry
- Lead Qualification

This allows different conversational workflows to be triggered depending on the customer's intent.

12. Lead Qualification

The chatbot is not only a support tool. It can also help qualify potential customers. For example, the assistant can identify information such as:

- Customer requirements
- Machine type
- Application
- Capacity requirements
- Attachment requirements
- Purchase intent
- Contact information

The conversation can therefore transition from:

> *Question → Product Discovery → Recommendation → Qualified Lead*

13. Lead Generation

When a customer demonstrates genuine interest, the AI can guide them toward an enquiry or contact workflow. Conceptually:

```
Customer Question
       |
Product Discovery
```

```
       |
Recommendation
       |
Customer Shows Interest
       |
Lead Information
       |
Business Follow-Up
```

This helps reduce the gap between customer support and sales.

14. Product Knowledge Integration

The chatbot was connected to the broader product ecosystem. This allows the AI experience to work alongside the website's product information. The assistant can help customers discover:

- Machines
- Product specifications
- Features
- Attachments
- Product differences
- Applications
- Related information

This makes the chatbot an extension of the website rather than an isolated AI widget.

15. Machine Recommendation Flow

A typical recommendation journey can look like:

```
Customer:
"I need a forklift for heavy warehouse work."
              |
AI identifies:
- Application
- Environment
- Capacity requirement
              |
Search Product Knowledge
              |
Find Relevant Machines
              |
Compare Available Options
              |
Recommend Suitable Product
              |
Show Product Information
              |
Continue to Enquiry
```

The AI therefore assists the customer throughout the discovery process.

16. FAQ Automation

Frequently asked questions can be handled automatically. Instead of requiring staff to answer repetitive questions, the AI can provide instant responses based on the company's knowledge base. This helps automate common support requests and allows human staff to focus on more complex customer requirements.

17. Knowledge-Grounded Responses

A major objective of the system was to reduce the risk of generic AI responses. The assistant is designed around business-specific knowledge rather than relying only on the model's general training.

```
Customer Question
       |
Retrieve Business Knowledge
       |
Provide Relevant Context
       |
LLM Generates Response
```

This makes responses more relevant to the actual products and services offered by the business.

18. AI Workflow Architecture

The overall architecture can be represented as:

```
                    CUSTOMER
                       |
                       v
                 CHAT INTERFACE
                       |
                       v
                 NODE.JS API
                       |
              +--------+--------+
              |                 |
              v                 v
        Intent Detection    Conversation
              |                 Context
              |
              v
        Semantic Search
              |
              v
       Knowledge Base
              |
              v
       Relevant Context
              |
              v
       AI / LLM Provider
       +------+------+
       |      |      |
       v      v      v
    OpenAI  Claude  Gemini
       |      |      |
       +------+------+
              |
              v
        AI Response
              |
              v
      Recommendation /
      Support / Lead
```

19. Backend Architecture

The backend was implemented using Node.js and Express. It acts as the orchestration layer between the customer-facing chatbot and AI services. Responsibilities include:

- Chat request handling
- Conversation management
- LLM communication
- Knowledge retrieval
- Product data retrieval
- AI response processing
- Lead information handling
- API integration
- Error handling

This keeps sensitive AI/API logic on the server rather than exposing it directly to the frontend.

20. MongoDB Conversation Management

MongoDB is used for application data and conversation-related information. The system can maintain conversation information such as:

```
Session
+-- Visitor Name
+-- Visitor Email
+-- Visitor Phone
+-- Selected Product
+-- Messages
     +-- User
     +-- AI
     +-- User
     +-- AI
```

This provides a structured way to maintain customer conversations and connect them with relevant products and leads.

21. Product Context

The chatbot can associate a conversation with a selected product. For example:

```
Customer opens:
QWF35 Product
        |
Opens AI Assistant
        |
Conversation automatically has
product context
        |
Customer asks:
"What attachments can I use?"
```

The assistant can then provide a more relevant response because it understands the product context.

22. AI-Powered Customer Journey

The complete customer journey can be summarized as:

```
Website Visitor
      |
Opens AI Assistant
      |
Asks Question
      |
AI Understands Requirement
      |
Searches Knowledge Base
      |
Retrieves Product Information
      |
Generates Contextual Response
      |
Recommends Product
      |
Answers Follow-Up Questions
      |
Qualifies Customer
      |
Generates Lead / Enquiry
```

23. Performance Impact

The implementation was designed to significantly reduce the time required for customers to receive initial support.

90%	5,000+	3
Reduction in support response time	Daily conversations supported	Major LLM providers integrated

The AI assistant provides immediate responses to common customer questions instead of requiring a human representative to manually respond to every initial enquiry.

Conversation Capacity

The system was designed to support 5,000+ conversations daily, providing scalable automated assistance for a large volume of customer interactions.

Multi-Provider AI

The platform supports 3 major LLM providers:

- OpenAI
- Anthropic Claude
- Google Gemini

Key Technical Challenges

Challenge 1 — Business-Specific Knowledge

Generic LLM responses were not sufficient for product-specific questions.

Solution: Implemented knowledge-base retrieval and RAG architecture to provide relevant business context to the model.

Challenge 2 — Product Recommendations

Customers often describe requirements rather than product names.

Solution: Used conversational context, semantic search, and product information to identify relevant products.

Challenge 3 — Multiple AI Providers

Different AI providers have different APIs and response formats.

Solution: Designed the backend AI integration layer to support multiple LLM providers while keeping the customer-facing interface consistent.

Challenge 4 — Maintaining Conversation Context

Customers naturally ask multiple related questions during one conversation.

Solution: Implemented session-based conversation management so previous messages and product context can be maintained.

Challenge 5 — Converting Support Into Leads

The chatbot needed to do more than answer questions.

Solution: Designed conversational workflows that can transition from support to product recommendation and eventually lead qualification.

AI Agent vs Traditional Chatbot

The major difference in the project is that it was not designed as a simple static FAQ chatbot.

Traditional Chatbot

```
Question
   |
Keyword Match
   |
Predefined Answer
```

AI Assistant

```
Question
   |
Understand Intent
   |
Understand Context
   |
Semantic Retrieval
   |
Knowledge Base
   |
LLM Reasoning
   |
```

```
Generate Response
    |
Recommend / Qualify / Assist
```

This makes the system substantially more flexible for real customer conversations.

My Contribution

I worked across the full AI application lifecycle, including:

- AI chatbot development
- AI agent workflows
- OpenAI integration
- Anthropic Claude integration
- Google Gemini integration
- RAG implementation
- Semantic search
- Vector embeddings
- Knowledge-base integration
- Product recommendation workflows
- Conversational context management
- Lead qualification
- Node.js / Express APIs
- MongoDB conversation storage
- Product context integration
- Backend AI orchestration
- Production deployment
- AI workflow debugging and optimization

Final Outcome

The result was an AI-powered customer assistance platform capable of supporting customers throughout the product discovery journey.

Instead of functioning as a simple question-and-answer bot, the system acts as an intelligent digital sales and support assistant. It combines:

> *LLMs + RAG + Semantic Search + Knowledge Base + Product Data + AI Agents + Conversation Context + Lead Qualification*

The platform provides immediate customer assistance, helps visitors discover relevant machinery, answers product questions, automates FAQs, and can transition qualified customer conversations into business leads.

Project Impact

90%	Reduction in support response time
5,000+	Daily conversations supported
3	Major LLM providers integrated
RAG + Semantic Search	Business-specific knowledge retrieval
AI Agents	Automated intelligent workflows
Lead Qualification	Support-to-sales automation